Characteristics and Classifications of living organisms

MRS GREN

Movement: An action by an organism or part of an organism causing a change of position or place

Respiration: The chemical reaction in cells that breaks down nutrient molecules and release energy for metabolism.

Glucose + Oxygen → Carbon dioxide + Water + energy

Sensitivity: The ability to detect and respond to changes in the internal or external environment

Growth: A permanent increase in size and dry mass

Reproduction: The process that makes more of the same kind of an organism

Excretion: The removal of waste products of metabolism and substances in excess of requirements

Nutrition: The taking in of materials for energy, growth, and development.

Species: A group of organisms that can reproduce to produce a fertile offspring.

Binomial system of naming: An internationally agreed system in which the scientific name of an organism is made up of two parts: Genus and species

Genus: Closely related species.

Genus specie

1- First letter of the genus should be capitalized

2- Should be underlined

3- Should be in Italics

Example: Humas → Homo sapiens

The Five Kingdoms

1- Animals
2- Plants
3- Prokaryotes
4- Fungi
5- Protochista

1- Animals : Characteristics shared by arthropods and vertebrates:

* Multicellular
* Have nucleus but no cell wall or chloroplast
* Feed on organic substances made by other living organisms:

Heterotrophic feeders

1- Animals
- Vertebrates : Have a backbone
- Invertebrates : Do not have a backbone
  - * Arthropods

* Invertebrates : Characteristics :

Shared by all arthropods :
- Segmented bodies
- Jointed legs
- Hard exoskeleton made out of chitin

1- Insects :
* Three pairs of legs
* Body divided into three parts :
  - Head
  - thorax
  - Abdomen
* One pair of antennae
* One pair of compound eyes

2- Arachnids :
* Four pairs of legs
* Body divided into two parts
  - Cephalothorax
  - Abdomen
* No antennae
* Several pairs of simple eyes
* A pair of pedipalps adapted for biting and poising the prey

3- Crustacea :
* Five pairs of legs
* Body divided into two parts
  - Cephalothorax
  - Abdomen
* Two pairs of antennae
* One pair of compound eyes
* Exoskeleton

4- Myriapods :
* Ten or more of legs
* Body not divided clearly to thorax and abdomen
* Segmented body
* One pair of antennae
* Simple eyes

* Vertebrates :

1- Mammals : * Have mammary glands
* Have sweat glands
* Body covered in fur
* Have four limbs
* Give birth
* Have external ears
* Warm blooded

2- Reptiles : * Body covered in dry scales
* lay eggs with leathary rubbary shells
* Cold blooded
* Four legs except snakes.

3- Fish : * Body covered in wet scales
* Have fins and streamlined bodies to resist water current
* Have gills to breath
* Cold blooded
* lay jelly-covered eggs in water

4- Amphibians : * Have moist skin
* lay jelly-covered eggs in water
* Tadpoles live in water and breath through gills. Adults live on land and breath through lungs and gills.

5- Birds : * Body covered with feathers
* Have 2 wings and 2 legs
* Have beaks
* lay eggs with hard shells
* legs are covered with scales

2- Plants: Characteristics

* Multicellular
* Have nucleus, cellulose cell walls, and chloroplasts.
* Can make their own food by photosynthesis: Autotrophic feeders
* May have roots, stems, leaves

Plants

Flowering

Monocotolydens | Dicotyledons

- Have roots, stems & leaves
- Reproduce by flowers and seeds
- Seeds are produced inside the ovary of the flower.

Non-Flowering

Have roots | No roots

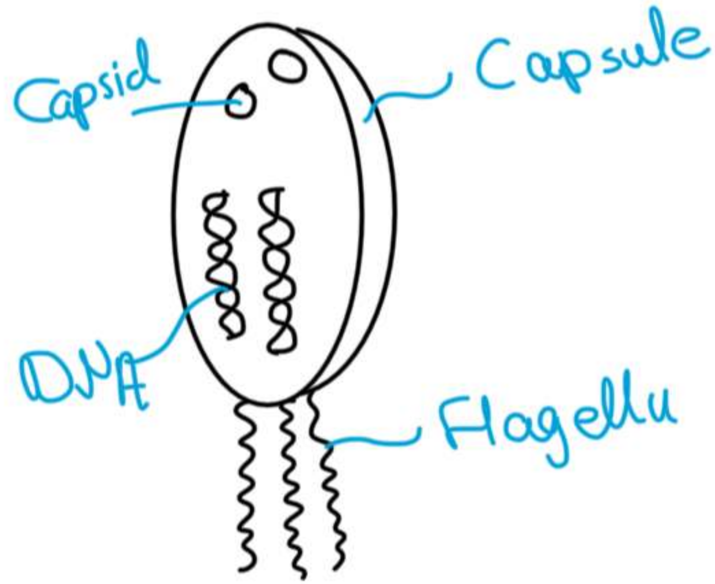
Ferns | Mosses

- Have roots, stems & leaves
- Reproduce by spores
- Do not produce flower

Monocotyledon	Dicotyledon
- One cotyledon	- Two cotyledons
- Long, narrow leaves with Parallel veins	- leaves are broad with netlike veins
- Vascular bundle is Complexly arranged	- Vascular bundle arranged in a ring
- Fibrous root system	- Taproot system
- Floral parts in multiples of three	- Floral parts in multiples of four or five

3- Prokaryotes : Bacteria

* Unicellular with a naked DNA in a loop
* Cellwalls are made of peptidoglycan or murien
* Have no mitochondria, no true nucleus and other membrane bound organelles
* Have plasmids which are extra pieces of DNA used in biotechnology
* May have a capsule
* Have flagella for movement



4- Fungi :

* Muticellular except for yeast which is unicellular
* Heterotrophic feeders, or saprotrophic feeders, which means they feed on dead organisms, or parasitic feeders, which means they feed on living organisms and harm them in the process.
* Have chitin cell walls.
* Have hyphae which extend from the main network called mycellium for feeding. They secret digestive enzymes for extracellular digestion, and absorbing digested nutrients.
* Reproduce by spores.

5- Protictista :

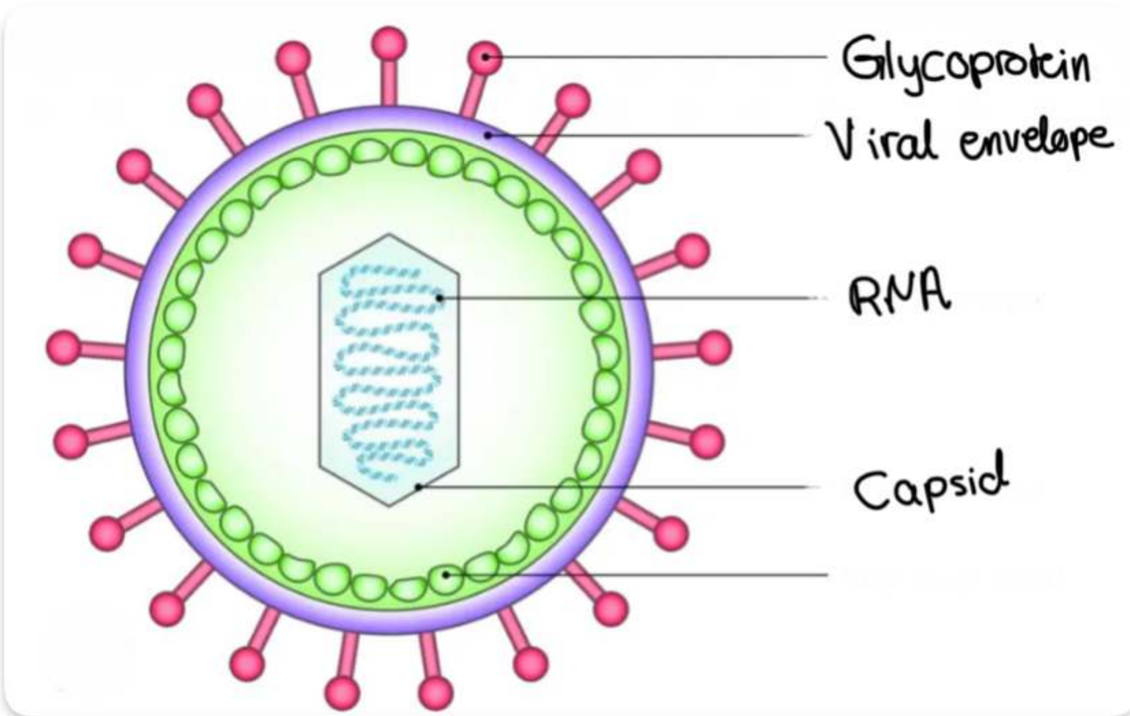
* Mostly unicellular, but some are multicellular
* Have cellulose cell walls and chloroplasts
* Some are autotrophic and some are heterotrophic feeders
* Have their chromosomes enclosed in a nuclear membrane to form a nucleus.

Viruses:

* They are not considered living things because: 1- They do not feed, respire, execrete or grow. 2- They do not have nucleus, cytoplasm, cell organelles, or cell membrane.

* They invade host cells, take over the cell's machinery to replicate. New viruses are released from the host cell and they go to invade new cells.

* A virus consists of genetic material (DNA or RNA) inside a protein coat: Capsid.



- DNA: Deoxyribo nuclic acid

- DNA makes up chromosomes and each piece of DNA consits of bases.

- 4 types of bases made of amino acids make up the DNA: A, C, G, T.

A pairs with T

C pairs with G

* Living organisms which share similar DNA sequence would be more closely related to each other

* Organisms that share similar DNA sequence may have evolved from a common ancestor millions of years ago